

Structural and Algorithmic Analysis of the Exact Minimum Fleet Size in the Flexible SPDP

Abstract

This note analyzes the exact minimum fleet size $k_{\min}^*$ in the flexible skip pick-up and delivery problem. It clarifies the underlying optimization problem, identifies a route-exactness issue in the current 2-index formulation, evaluates the duration-based explanation in Sections 5 and 6, establishes complexity and structural results, and develops a matching-aware lower-bound hierarchy and an exact logic-based Benders algorithm. The main conclusion is that the observed computational ease of $k_{\min}^*$ is not a general tractability phenomenon. It is currently explained by a lower-bound–feasible-solution sandwich that closes at the first tested fleet size on the benchmark instances. The most promising new contribution is to exploit the latent one-to-one matching between picked-up skips and same-type deliveries before solving the route-sequencing problem.

1 Status convention and standing assumptions

Every proposed result is marked as one of the following:

- (i) rigorously established from the manuscript’s assumptions;
- (ii) a conjecture or missing result requiring proof;
- (iii) a plausible research direction; or
- (iv) an empirical claim requiring computational validation.

Unless otherwise stated, the analysis concerns the physical route problem described in Section 2 of the manuscript. Results involving shortcutting or direct travel-time lower bounds additionally assume that the travel-time matrix satisfies the directed triangle inequality. Complexity statements assume rational input data encoded in binary.

2 Definition and standalone formulations of the minimum-fleet problem

Let $\mathcal{R}_T$ denote the family of all elementary open routes that start empty with a pick-up action, end empty with a delivery action, respect the two-skip vehicle-state transitions, empty every collected full skip at its designated treatment location, and have duration at most T . For $q \in \mathcal{R}_T$ and action $v \in \mathcal{C}^p \cup \mathcal{C}^d$, let $a_{vq} = 1$ if route q serves action v and let $a_{vq} = 0$ otherwise.

Definition 1 (Exact minimum fleet size). *The exact minimum fleet size is*

$$k_{\min}^* = \min_{Q \subseteq \mathcal{R}_T} \left\{ |Q| : \sum_{q \in Q} a_{vq} = 1 \quad v \in \mathcal{C}^p \cup \mathcal{C}^d \right\}. \quad (1)$$

Thus, $k_{\min}^*$ is the minimum number of duration-feasible vehicle routes required to perform every pick-up and delivery action exactly once. It is independent of the vehicle fixed charge and of the variable travel-cost coefficients.

Definition 2 (Decision problem FLEET-SPDP(k)). *Given an SPDP instance and an integer k , determine whether there exists a feasible route set $Q \subseteq \mathcal{R}_T$ such that $|Q| \leq k$.*

The at-most- k formulation is preferable for algorithmic purposes because its feasibility predicate is monotone in k .

2.1 Set-partitioning formulation

Status: rigorously established from the manuscript's assumptions.

An exact route formulation is

$$\min \sum_{q \in \mathcal{R}_T} x_q \quad (2)$$

$$\text{s.t.} \quad \sum_{q \in \mathcal{R}_T} a_{vq} x_q = 1 \quad v \in \mathcal{C}^p \cup \mathcal{C}^d, \quad (3)$$

$$x_q \in \{0, 1\} \quad q \in \mathcal{R}_T. \quad (4)$$

Consequently, computing $k_{\min}^*$ is a minimum-cardinality resource-constrained path-cover or set-partitioning problem on the state-expanded multigraph.

An equivalent threshold formulation removes the duration limit temporarily and defines

$$M(k) = \min_{\substack{Q \text{ route-feasible except for } T \\ |Q| \leq k}} \max_{q \in Q} D(q). \quad (5)$$

Then

$$k_{\min}^* = \min\{k : M(k) \leq T\}. \quad (6)$$

3 A foundational route-exactness issue

The manuscript defines

$$k_{\min}^* = \min\{K(y) : (y, B) \in \mathcal{F}^{\text{IP}}\}. \quad (7)$$

However, only nonnegative travel and service times are assumed. The degree and state-flow equations can therefore admit a disconnected customer-action cycle of zero duration.

For example, consider one request with co-located pick-up, treatment, and delivery and with all travel and ~~service times equal to zero~~. Selecting an internal cycle

$$C_i^p \longrightarrow C_i^d \longrightarrow C_i^p$$

can satisfy the action-degree, state-consistency, and time-potential constraints while having $K(y) = 0$. The physical problem nevertheless requires one route.

Theorem 1 (Route decomposition; statement to be added). *Suppose that:*

- (a) *no selected edge enters the start node or leaves the end node;*
- (b) *every selected departure enters a pick-up action in the empty state;*
- (c) *every selected return leaves a delivery action in the empty state;*
- (d) *every directed cycle consisting only of customer-action nodes has strictly positive total duration; and* by saving time
- (e) *the multigraph construction and every graph-reduction rule preserve complete route realizability in both directions.*

Then every integer point of $\mathcal{F}^{\text{IP}}$ decomposes into exactly $K(y)$ state-compatible start-to-end paths, each of duration at most T . Conversely, every feasible physical route set maps to an integer point of $\mathcal{F}^{\text{IP}}$.

Status: a conjecture or missing result requiring proof.

The proof should first use the action-degree equations to show that each selected connected component is a path or a cycle. The time-potential inequalities and assumption (d) eliminate action-only cycles. State balance then joins edges only through compatible post-service states, and the boundary time constraints impose the duration limit on every remaining path.

The manuscript should either prove Theorem 1, **impose strictly positive action service times**, or add explicit connectivity/subtour-elimination constraints. The current state-compression propositions prove cardinality reduction but do not by themselves prove route-level lifting after all symmetry and dominance filters.

4 Assessment of the current duration-based explanation

Recall

$$K(y) = \sum_{\substack{e \in \mathcal{E}: u_e=0 \\ v_e \in \mathcal{C}^p}} y_e, \quad D(y) = \sum_{e \in \mathcal{E}} t_e y_e, \quad (8)$$

$$H^* = \min\{D(y) : (y, B) \in \mathcal{F}^{\text{IP}}\}. \quad (9)$$

4.1 What is already valid

Proposition 1 (Duration route bound). *For any valid lower bound $\underline{H} \leq H^*$,*

$$k_{\min}^* \geq \left\lceil \frac{\underline{H}}{T} \right\rceil. \quad (10)$$

Status: rigorously established from the manuscript's assumptions.

For any K -route feasible solution,

$$\underline{H} \leq H^* \leq D(y) \leq TK.$$

The route-decomposition condition discussed above is needed for the final inequality.

The Wøhlk–Laporte workload dominance argument is also valid under the directed triangle inequality, provided that the proof explicitly introduces a token-to-delivery matching and observes that each physical travel leg is charged to at most two onboard skips.

The Introduction should not state that dividing the duration optimum by T computes the minimum fleet size. It computes only a lower bound. A clearer notation is

$$L_{\text{dur}} = \left\lceil \frac{\max\{\underline{H}, H^{\text{WL}}\}}{T} \right\rceil,$$

reserving $k_{\min}^*$ for the exact fleet minimum.

4.2 The actual certificate behind the computational results

Theorem 2 (Fleet-size sandwich certificate). *Let L be any certified lower bound satisfying $L \leq k_{\min}^*$. If a feasible route set with exactly L routes is found, then $k_{\min}^* = L$.*

Status: rigorously established from the manuscript’s assumptions.

More generally, if a feasible solution uses $\bar{k}$ routes and a certified duration lower bound satisfies

$$\underline{H} > (\bar{k} - 1)T, \tag{11}$$

then $k_{\min}^* = \bar{k}$. Thus the duration IP need not be solved to optimality; its certified lower bound only needs to cross the relevant threshold.

The direct explanation of the reported computational ease is therefore that the duration-derived lower certificate and a feasible fleet-size certificate meet at the first candidate for all tested instances. This is a strong empirical property of the benchmark distribution, not a worst-case tractability result.

Status: an empirical claim requiring computational validation.

The claim that the initial duration-based lower bound is usually exact should be evaluated on additional instance classes specifically designed to stress workload packing and route consolidation.

4.3 What Section 6 does and does not explain

The empty-state splitting proposition correctly explains why total-duration and travel-cost objectives may weakly favor a larger number of separated routes. It does not explain why minimum-fleet optimization should be easy: minimizing K has the opposite consolidation incentive and retains the route-packing layer removed by splitting.

The statement

$$\phi_{\text{LP}}(k_{\min}^*) \leq c_{\min}$$

is a possible empirical property, not a consequence of relaxation. The equality-constrained LP defining $\phi_{\text{LP}}(k_{\min}^*)$ should be solved directly. A final branch-and-bound lower bound equal to $f k_{\min}^* + c_{\min}$ does not by itself exhibit a fractional solution attaining both components simultaneously.

5 Complexity results and counterexamples

5.1 NP-completeness when one vehicle is requested

Theorem 3 (One-route feasibility). *Deciding whether $k_{\min}^* \leq 1$ is NP-complete even with one skip type, symmetric metric travel times in $\{0, 1, 2\}$, positive homogeneous service times, and $\alpha_i = \beta_i = \omega_i$ for every request.*

Status: rigorously established from the manuscript’s assumptions.

Proof. Membership in NP follows by checking an action sequence and its treatment operations. For hardness, reduce from Hamiltonian Path. Given an undirected graph $G = (V, E)$ with $n = |V|$, create one request at every vertex and set

$$t_{uv} = \begin{cases} 0, & u = v, \\ 1, & \{u, v\} \in E, \\ 2, & \{u, v\} \notin E. \end{cases}$$

This is a metric. Set $s^p = s^e = s^d = 1$ and $T = 4n - 1$. A Hamiltonian path produces one route by completing each request locally and visiting the vertices in path order. Conversely, any one-route solution uses exactly $3n$ units of service time and at most $n - 1$ units of travel time. Visiting n distinct locations requires at least $n - 1$ positive travel legs. Hence every such leg must have length one, and their order defines a Hamiltonian path in G . $\square$

Theorem 3 implies that no exact polynomial or general pseudo-polynomial algorithm exists unless $P = NP$. It also shows that the problem is para-NP-hard in the fleet-size parameter k .

Moreover, any $(2 - \varepsilon)$ -approximation for the integer fleet objective would distinguish instances with optimum one from those with optimum at least two. Therefore, no such approximation exists unless $P = NP$.

5.2 Exact embedding of bin packing

Theorem 4 (Star-metric bin-packing subclass). *There is a metric subclass of the flexible SPDP in which computing $k_{\min}^*$ is exactly a one-dimensional bin-packing problem.*

Status: rigorously established from the manuscript’s assumptions.

Proof. Place every pick-up and delivery at a common hub o . Let request i have treatment location ℓ_i , where the travel metric is a star:

$$t_{o, \ell_i} = a_i, \quad t_{\ell_i, \ell_j} = a_i + a_j.$$

Use one skip type and set $s^p = s^e = s^d = 1$. A route serving request set S must traverse every leaf edge associated with S twice and incurs three service units per request. Its minimum duration is therefore exactly

$$\sum_{i \in S} (2a_i + 3), \quad (12)$$

which is attainable by sequential service. Hence the requests are bin-packing items of sizes $2a_i + 3$ and route duration T is the bin capacity. $\square$

For a 3-PARTITION instance satisfying

$$\frac{B}{4} < a_i < \frac{B}{2}, \quad \sum_{i=1}^{3m} a_i = mB,$$

set $T = 2B + 9$. Four requests cannot fit on one route, the total workload is mT , and a packing into m routes exists if and only if the integers admit a 3-partition. Thus the problem is strongly NP-hard even with one customer location, one skip type, positive services, metric travel, and at most three requests on a route.

This construction also has

$$H^* = mT, \quad \left\lceil \frac{H^*}{T} \right\rceil = m.$$

Consequently, deciding feasibility at an exact duration lower bound remains strongly NP-hard.

5.3 Arbitrarily weak duration bound

Proposition 2 (Unbounded weakness of the duration rounding). *There is no constant worst-case relationship between $\lceil H^*/T \rceil$ and $k_{\min}^*$, even with metric travel times and positive homogeneous service times.*

Status: rigorously established from the manuscript's assumptions.

Proof. For any n , create n distinct sites ℓ_i and let

$$\alpha_i = \beta_i = \omega_i = \ell_i, \quad s^p = s^e = s^d = 1.$$

Set $T = 3n$ and let the distance between any two distinct sites be $3n + 1$. This is a metric. Singleton routes give $H^* = 3n$, so $\lceil H^*/T \rceil = 1$. However, no route can visit two distinct sites, and therefore $k_{\min}^* = n$. $\square$

6 Structural results specific to the fleet objective

Let

$$S = s^p + s^e + s^d, \quad \mathcal{I}_h = \{i \in \mathcal{I} : h_i = h\}.$$

For $i, j \in \mathcal{I}_h$, define the direct singleton duration

$$d_{ij} = S + t_{\alpha_i, \beta_i} + t_{\beta_i, \omega_j}. \quad (13)$$

6.1 Polynomial unlimited-fleet feasibility test

Theorem 5 (Typewise perfect-matching characterization). *Assume the directed triangle inequality. The SPDP is feasible with an unlimited fleet if and only if, for every type h , the bipartite graph*

$$(i, j) \in E_h \iff i, j \in \mathcal{I}_h, \quad d_{ij} \leq T \quad (14)$$

contains a perfect matching.

Status: rigorously established under the directed triangle inequality.

Proof. Label every physical skip at its pick-up and preserve the label after emptying. Every feasible route solution induces, separately for each type, a bijection from picked-up full skips to the delivery actions consuming the corresponding empty skips. If token i is delivered at j , shortcutting its ridden path gives

$$d_{ij} \leq D(q) \leq T.$$

Thus the induced typewise matching uses only edges in E_h . Conversely, a perfect matching gives one feasible singleton route $i \rightarrow \beta_i \rightarrow j$ for every matched pair. $\square$

Theorem 5 gives a polynomial infeasibility test and a certified n -route solution whenever the instance is feasible. The total matching complexity is

$$O\left(\sum_{h \in \mathcal{H}} |\mathcal{I}_h|^3\right) \leq O(n^3).$$

6.2 Contiguous feasible fleet sizes

Theorem 6 (Fleet refinement). *Under the directed triangle inequality, if a feasible k -route solution exists, then feasible solutions exist for every fleet size $k, k+1, \dots, n$.*

Status: rigorously established under the directed triangle inequality.

Proof. If $k < n$, some route contains at least two labeled token lifecycles. Select one token matched from pick-up i to delivery j . Create a singleton route $i \rightarrow \beta_i \rightarrow j$ and delete its pick-up, treatment, and delivery operations from the original route. Shortcutting the residual route cannot increase its duration or capacity usage. Repetition gives all larger fleet sizes through n . $\square$

Hence the feasible exact fleet sizes are precisely

$$\{k_{\min}^*, k_{\min}^* + 1, \dots, n\}.$$

This justifies both sequential equality testing and binary search, although the at-most- k decision problem remains the conceptually cleaner oracle.

6.3 Fleet-specific graph reductions

Proposition 3 (Minimum-duration parallel-edge dominance). *For fleet feasibility or duration minimization, among all parallel edges with identical ordered endpoints and identical start and end states, it is sufficient to retain one minimum-duration edge.*

Status: rigorously established from the manuscript’s assumptions.

Replacing any other edge by a minimum-duration representative preserves all state transitions and action coverage and weakly reduces the route duration. The travel-cost dimension is irrelevant for the auxiliary fleet problem. This rule is therefore stronger than the time-cost Pareto filter needed for the original objective.

Let $\delta^0(u, \sigma)$ and $\delta^d(v, \sigma')$ be shortest-duration prefix and suffix lower bounds in the node-state graph. The edge $e = (u, v)$ can also be removed when

$$\delta^0(u, \sigma(u)) + t_e + \delta^d(v, \sigma(v)) > T. \quad (15)$$

This is another exact duration-specific preprocessing rule.

6.4 Busy-block decomposition

Definition 3 (Busy block). *A busy block is an empty-to-empty route segment that begins with a pick-up, ends with a delivery, and has no internal customer action after which the vehicle is empty.*

Proposition 4 (Fully split normal form). *Under the boundary-arc assumptions of the manuscript’s splitting proposition, there exists a total-duration-optimal solution consisting only of busy blocks. The same is true for travel-cost-only minimization.*

Status: rigorously established from the manuscript’s assumptions.

Repeatedly split every internal empty-state delivery-to-pick-up connector. Each split weakly decreases the relevant objective, and the process terminates after finitely many splits. Consequently, these connectors can be deleted exactly from the H^* auxiliary model. They cannot be deleted from the minimum-fleet model because joining busy blocks is precisely the hard consolidation and sequence-dependent packing layer.

For a fixed set of busy blocks $\mathcal{B}$ and selected connectors X , one has

$$K = |\mathcal{B}| - |X|, \quad (16)$$

provided the selected connectors form vertex-disjoint chains. The original cost becomes

$$\sum_{b \in \mathcal{B}} C(b) + f|\mathcal{B}| + \sum_{(b, b') \in X} (c_{bb'} - f), \quad (17)$$

which formalizes the split-versus-merge trade-off in Section 6.

6.5 Absence of a matroid exchange structure

Proposition 5 (No general matroid structure). *For a fixed token matching, the family of token sets that fit on one route is hereditary under deletion and shortcutting, but need not satisfy the matroid exchange axiom.*

Status: rigorously established under the directed triangle inequality.

The star/bin-packing subclass with $T = 12$ and token durations 8, 6, 6 has feasible sets $A = \{8\}$ and $B = \{6, 6\}$. Although $|A| < |B|$, neither member of B can augment A . Thus a general greedy or exchange-based exact algorithm should not be expected.

7 A matching-aware lower-bound hierarchy

For $i, j \in \mathcal{I}_h$, define

$$w_{ij} = S + \frac{1}{2} (t_{\alpha_i, \beta_i} + t_{\beta_i, \omega_j}), \quad (18)$$

and retain only the singleton-feasible matching edges $d_{ij} \leq T$.

7.1 Routewise matching workload

Theorem 7 (Routewise matching-workload inequality). *Assume the directed triangle inequality. Every feasible route q induces, for each type h , a bijection $\mu_{q,h}$ from the type- h pick-ups on q to the type- h deliveries on q such that*

$$\sum_{h \in \mathcal{H}} \sum_{i \in P_h(q)} w_{i, \mu_{q,h}(i)} \leq D(q) \leq T. \quad (19)$$

Status: rigorously established under the directed triangle inequality.

Proof. Conceptually label every full skip at pick-up. Emptying changes the token from full to empty but preserves its label, and the label is assigned to the delivery that consumes the empty token. Let $L_{ij}(q)$ be the vehicle travel accumulated while token i , eventually delivered at j , is aboard. Directed triangle inequality gives

$$t_{\alpha_i, \beta_i} + t_{\beta_i, \omega_j} \leq L_{ij}(q).$$

Every travel segment is charged to at most two onboard tokens, and therefore

$$\sum_{(i,j) \in \mu_q} L_{ij}(q) \leq 2V(q),$$

where $V(q)$ is the route travel duration. If the route contains r token pairs, it incurs exactly rS service time. It follows that

$$\sum_{(i,j) \in \mu_q} w_{ij} \leq rS + V(q) = D(q).$$

□

7.2 Polynomial assignment bound

Define

$$H^{\text{match}} = \sum_{h \in \mathcal{H}} \min_{M_h} \sum_{(i,j) \in M_h} w_{ij}, \quad (20)$$

where M_h ranges over perfect matchings in the singleton-feasible graph for type h .

Proposition 6 (Matching-workload hierarchy). *Under the directed triangle inequality,*

$$H^{\text{WL}} \leq H^{\text{match}} \leq H^*. \quad (21)$$

Status: rigorously established under the directed triangle inequality.

The first inequality follows because the assignment problem cannot repeatedly select the same cheapest delivery candidate, whereas H^{WL} uses independent row minima. The second follows by taking the union of the matchings induced by the routes of any feasible solution and applying Theorem 7.

Consequently,

$$L_{\text{match}} = \left\lceil \frac{H^{\text{match}}}{T} \right\rceil \quad (22)$$

is polynomially computable and dominates the Wøhlk–Laporte analytical route bound.

A second inexpensive bound uses the fact that every route serving r token pairs incurs rS service time. If

$$q = \left\lfloor \frac{T}{S} \right\rfloor,$$

then

$$k_{\min}^* \geq \left\lceil \frac{n}{q} \right\rceil, \quad (23)$$

provided $S > 0$ and $q \geq 1$. This may be strictly stronger than $\lceil nS/T \rceil$ because it recognizes request indivisibility.

7.3 Explicit dual-feasible-function assignment bounds

Definition 4 (Dual-feasible function). *A function $g : [0, 1] \rightarrow [0, 1]$ is dual feasible if, for every finite sequence $x_1, \dots, x_r \in [0, 1]$,*

$$\sum_{\ell=1}^r x_\ell \leq 1 \implies \sum_{\ell=1}^r g(x_\ell) \leq 1. \quad (24)$$

For a dual-feasible function g , define

$$A_g = \sum_{h \in \mathcal{H}} \min_{M_h} \sum_{(i,j) \in M_h} g\left(\frac{w_{ij}}{T}\right), \quad (25)$$

where M_h ranges over the perfect matchings in the singleton-feasible bipartite graph for type h .

Proposition 7 (Dual-feasible matching bound). *Assume the directed triangle inequality. For every dual-feasible function g ,*

$$k_{\min}^* \geq L_g := \lceil A_g \rceil. \quad (26)$$

Status: rigorously established under the directed triangle inequality.

Proof. Consider any feasible solution using k routes. By Theorem 7, every route q induces a matching μ_q satisfying

$$\sum_{(i,j) \in \mu_q} \frac{w_{ij}}{T} \leq 1.$$

Dual feasibility of g therefore implies

$$\sum_{(i,j) \in \mu_q} g\left(\frac{w_{ij}}{T}\right) \leq 1.$$

The union of the route-induced matchings is a typewise perfect matching. Denoting this matching for type h by $\widehat{M}_h$, we obtain

$$\begin{aligned} A_g &\leq \sum_{h \in \mathcal{H}} \sum_{(i,j) \in \widehat{M}_h} g\left(\frac{w_{ij}}{T}\right) \\ &= \sum_{q=1}^k \sum_{(i,j) \in \mu_q} g\left(\frac{w_{ij}}{T}\right) \leq k. \end{aligned}$$

Since this holds for every feasible solution, $A_g \leq k_{\min}^*$. Because $k_{\min}^*$ is integral, $\lceil A_g \rceil \leq k_{\min}^*$. $\square$

The preceding bound becomes operational once an explicit dual-feasible function is selected. The following rounding family is particularly suitable because it detects token weights lying just above the thresholds $1/2, 1/3, \dots$

Proposition 8 (Rounding dual-feasible functions). *For every integer $m \geq 2$, define $g_m : [0, 1] \rightarrow [0, 1]$ by*

$$g_m(x) = \begin{cases} 0, & x = 0, \\ \frac{\lceil mx \rceil - 1}{m - 1}, & x \in (0, 1]. \end{cases} \quad (27)$$

Then g_m is dual feasible.

Status: rigorously established from the manuscript's assumptions.

Proof. Let $x_1, \dots, x_r \in [0, 1]$ satisfy

$$\sum_{\ell=1}^r x_\ell \leq 1.$$

Terms with $x_\ell = 0$ make no contribution, so consider only the positive terms and define

$$a_\ell := \lceil mx_\ell \rceil - 1.$$

Each a_ℓ is a nonnegative integer and satisfies

$$a_\ell < mx_\ell.$$

Consequently,

$$\sum_{\ell=1}^r a_{\ell} < m \sum_{\ell=1}^r x_{\ell} \leq m.$$

Since $\sum_{\ell=1}^r a_{\ell}$ is an integer, it follows that

$$\sum_{\ell=1}^r a_{\ell} \leq m - 1.$$

Therefore,

$$\sum_{\ell=1}^r g_m(x_{\ell}) = \frac{1}{m-1} \sum_{\ell=1}^r a_{\ell} \leq 1,$$

which proves dual feasibility. $\square$

For example, the first two members of this family are

$$g_2(x) = \begin{cases} 0, & 0 \leq x \leq \frac{1}{2}, \\ 1, & \frac{1}{2} < x \leq 1, \end{cases} \quad (28)$$

and

$$g_3(x) = \begin{cases} 0, & 0 \leq x \leq \frac{1}{3}, \\ \frac{1}{2}, & \frac{1}{3} < x \leq \frac{2}{3}, \\ 1, & \frac{2}{3} < x \leq 1. \end{cases} \quad (29)$$

Thus, g_2 counts every token whose matching workload exceeds half of a route as one whole route. Similarly, g_3 assigns one-half unit to a token exceeding one third of a route and one whole unit to a token exceeding two thirds of a route.

For $m \geq 2$, define

$$A_m := A_{g_m} = \sum_{h \in \mathcal{H}} \min_{M_h} \sum_{(i,j) \in M_h} g_m\left(\frac{w_{ij}}{T}\right) \quad (30)$$

and

$$L_m := \lceil A_m \rceil. \quad (31)$$

Propositions 7 and 8 immediately give

$$k_{\min}^* \geq L_m \quad \text{for every integer } m \geq 2. \quad (32)$$

Corollary 1 (Recovery of the service-cardinality bound). *Suppose $S > 0$ and let*

$$q = \left\lfloor \frac{T}{S} \right\rfloor \geq 1.$$

Then the rounding function g_{q+1} satisfies

$$L_{q+1} \geq \left\lceil \frac{n}{q} \right\rceil. \quad (33)$$

Consequently, L_{q+1} recovers, and may strictly strengthen, the service-cardinality bound (23).

Status: rigorously established under the directed triangle inequality.

Proof. Every retained matching edge satisfies $w_{ij} \geq S$. Moreover, the definition of q implies

$$\frac{S}{T} > \frac{1}{q+1}.$$

Hence

$$(q+1)\frac{w_{ij}}{T} \geq (q+1)\frac{S}{T} > 1,$$

so that

$$\left\lceil (q+1)\frac{w_{ij}}{T} \right\rceil \geq 2.$$

It follows from (27) that

$$g_{q+1}\left(\frac{w_{ij}}{T}\right) \geq \frac{1}{q}$$

for every retained matching edge. A collection of typewise perfect matchings contains exactly n edges in total. Therefore,

$$A_{q+1} \geq \frac{n}{q},$$

and taking ceilings proves (33). $\square$

The identity function

$$g_{\text{id}}(x) = x$$

is also dual feasible and recovers the assignment bound L_{match} . For any finite parameter library $\mathcal{P} \subseteq \mathbb{Z}_{\geq 2}$, we can therefore define the composite bound

$$L_{\text{DFF}} = \max \left\{ L_{\text{match}}, \max_{m \in \mathcal{P}} L_m \right\}. \quad (34)$$

Status: rigorously established under the directed triangle inequality.

For each m , the transformed edge costs are

$$c_{ij}^{(m)} = g_m\left(\frac{w_{ij}}{T}\right), \quad (35)$$

so L_m requires one minimum-cost perfect matching problem per request type. The complete bound (34) can be computed in

$$O\left((|\mathcal{P}|+1) \sum_{h \in \mathcal{H}} |\mathcal{I}_h|^3\right) \quad (36)$$

time and

$$O\left(\sum_{h \in \mathcal{H}} |\mathcal{I}_h|^2\right)$$

working space when the assignment problems are solved sequentially.

Status: a plausible research direction.

A practical initial library is

$$\mathcal{P} = \{2, \dots, \min\{n+1, 20\}\} \cup \{q+1\}, \quad (37)$$

with the term $q+1$ included when $S > 0$ and $q \geq 1$. The maximum must be taken over the resulting lower bounds; the bounds cannot be added because they are alternative relaxations of the same fleet requirement.

The rounding functions can be substantially stronger than the identity function near cardinality thresholds. For example, three unavoidable normalized matching weights equal to 0.51 contribute only

$$\lceil 3(0.51) \rceil = 2$$

under the identity function, but contribute

$$\lceil 3g_2(0.51) \rceil = 3$$

under g_2 . Similarly, five unavoidable weights equal to 0.34 give

$$\lceil 5(0.34) \rceil = 2$$

under the identity function and

$$\lceil 5g_3(0.34) \rceil = \left\lceil \frac{5}{2} \right\rceil = 3$$

under g_3 .

7.4 Matching-bin-packing relaxation

Let $x_{ijb} = 1$ if token i is matched to delivery j and assigned to bin b , and let $z_b = 1$ if bin b is used. Consider

$$L_{\text{MBP}} = \min \sum_{b=1}^n z_b \quad (38)$$

$$\text{s.t.} \quad \sum_{b=1}^n \sum_{j \in \mathcal{I}_{h_i}} x_{ijb} = 1 \quad i \in \mathcal{I}, \quad (39)$$

$$\sum_{b=1}^n \sum_{i \in \mathcal{I}_{h_j}} x_{ijb} = 1 \quad j \in \mathcal{I}, \quad (40)$$

$$\sum_{h \in \mathcal{H}} \sum_{i, j \in \mathcal{I}_h} w_{ij} x_{ijb} \leq T z_b \quad b = 1, \dots, n, \quad (41)$$

$$x_{ijb} \leq z_b \quad i, j, b, \quad (42)$$

$$x_{ijb}, z_b \in \{0, 1\}. \quad (43)$$

Only singleton-feasible pairs $d_{ij} \leq T$ are included.

For use below, let

$$E^{\mathcal{M}} = \bigcup_{h \in \mathcal{H}} E_h, \quad \mathfrak{M} = \prod_{h \in \mathcal{H}} \mathcal{M}_h, \quad s_{ij} = \frac{w_{ij}}{T}, \quad (44)$$

where $\mathcal{M}_h$ is the family of perfect matchings in E_h . For $M \in \mathfrak{M}$, let $\text{BP}(M)$ denote the optimal ordinary one-dimensional bin-packing value for the fixed item list $\{w_{ij} : (i, j) \in M\}$ and capacity T . The MBP optimum has the equivalent representation

$$L_{\text{MBP}} = \min_{M \in \mathfrak{M}} \text{BP}(M). \quad (45)$$

Indeed, every MBP solution selects a typewise perfect matching and packs its selected edge weights, while every typewise perfect matching and its bin packing define a feasible MBP solution. Thus an ordinary bin-packing algorithm solves the inner problem only after the matching has been fixed.

Status: rigorously established from the manuscript's assumptions.

7.4.1 Dual-feasible inequalities for the MBP formulation

Proposition 9 (Per-bin dual-feasible inequalities). *For every dual-feasible function g and every bin b , the inequality*

$$\sum_{h \in \mathcal{H}} \sum_{(i,j) \in E_h} g(s_{ij}) x_{ijb} \leq z_b \quad (46)$$

is valid for the integer MBP formulation.

Status: rigorously established from the manuscript's assumptions.

Proof. If $z_b = 0$, the linking constraints imply $x_{ijb} = 0$ for every (i, j) . If $z_b = 1$, the selected edges in bin b satisfy

$$\sum_{h \in \mathcal{H}} \sum_{(i,j) \in E_h} s_{ij} x_{ijb} \leq 1$$

by the capacity constraint. Dual feasibility of g then gives (46). $\square$

Let

$$y_{ij} = \sum_{b=1}^n x_{ijb}. \quad (47)$$

Summing (46) over the bins gives the aggregate inequality

$$\sum_{h \in \mathcal{H}} \sum_{(i,j) \in E_h} g(s_{ij}) y_{ij} \leq \sum_{b=1}^n z_b. \quad (48)$$

The vector y is the incidence vector of a typewise perfect matching. Minimizing the left-hand side of (48) over all such matchings recovers A_g in (25). Consequently,

$$L_g \leq L_{\text{MBP}} \quad \text{for every dual-feasible function } g. \quad (49)$$

The per-bin inequalities are redundant for integer capacity-feasible solutions, but they need not be redundant for the LP relaxation of the compact MBP formulation.

Status: an empirical claim requiring computational validation.

The extent to which (46) reduces the root gap, branch-and-bound nodes, and solution time must be evaluated computationally.

7.4.2 Separate, mixed, and common-matching DFF bounds

Let

$$\mathcal{G} = \{g^1, \dots, g^p\}$$

be a finite library of dual-feasible functions, and define

$$C_r(M) = \sum_{(i,j) \in M} g^r(s_{ij}), \quad r = 1, \dots, p. \quad (50)$$

Testing the functions independently gives the unrounded bound

$$D_{\text{sep}} = \max_{r=1, \dots, p} \min_{M \in \mathfrak{M}} C_r(M). \quad (51)$$

The associated integer fleet bound is $L_{\text{sep}} = \lceil D_{\text{sep}} \rceil$.

A stronger polynomial bound can be obtained by optimizing over convex combinations of the selected functions. Let

$$\Delta_p = \left\{ \lambda \in \mathbb{R}_+^p : \sum_{r=1}^p \lambda_r = 1 \right\}$$

and define

$$D_{\text{mix}} = \max_{\lambda \in \Delta_p} \min_{M \in \mathfrak{M}} \sum_{r=1}^p \lambda_r C_r(M). \quad (52)$$

For every $\lambda \in \Delta_p$, the function

$$g_\lambda(x) = \sum_{r=1}^p \lambda_r g^r(x) \quad (53)$$

is dual feasible. Hence D_{mix} is the best assignment bound generated by the convex hull of the chosen DFF library.

Finally, requiring one integral matching to be simultaneously evaluated by every function gives

$$D_{\text{rob}} = \min_{M \in \mathfrak{M}} \max_{r=1, \dots, p} C_r(M). \quad (54)$$

This common-matching bound is potentially stronger than D_{mix} , but it is not an ordinary linear assignment problem.

Proposition 10 (Optimized DFF-mixture formulation). *The value D_{mix} is the optimum of the linear program*

$$\min \quad \eta \quad (55)$$

$$\begin{aligned} s.t. \quad & \sum_{j: (i,j) \in E_h} y_{ij} = 1 & h \in \mathcal{H}, \quad i \in \mathcal{I}_h, \end{aligned} \quad (56)$$

$$\sum_{i:(i,j) \in E_h} y_{ij} = 1 \quad h \in \mathcal{H}, j \in \mathcal{I}_h, \quad (57)$$

$$\sum_{h \in \mathcal{H}} \sum_{(i,j) \in E_h} g^r(s_{ij}) y_{ij} \leq \eta \quad r = 1, \dots, p, \quad (58)$$

$$y_{ij} \geq 0 \quad h \in \mathcal{H}, (i,j) \in E_h. \quad (59)$$

Moreover, replacing (59) by $y_{ij} \in \{0, 1\}$ gives a compact formulation of D_{rob} .

Status: rigorously established from the manuscript's assumptions.

Proof. The feasible region defined by (56)–(59) without (58) is the product of the typewise bipartite perfect-matching polytopes and is therefore the convex hull of $\mathfrak{M}$. Applying the finite-dimensional minimax theorem to the bilinear function

$$(y, \lambda) \mapsto \sum_{r=1}^p \lambda_r \sum_{h \in \mathcal{H}} \sum_{(i,j) \in E_h} g^r(s_{ij}) y_{ij}$$

shows that (52) equals the minimum, over this matching polytope, of the maximum of the p transformed matching loads. Introducing the epigraph variable η yields (55)–(59). Imposing integrality restricts the matching vector to $\mathfrak{M}$ and therefore yields (54). $\square$

Proposition 11 (DFF–MBP hierarchy). *Suppose the identity function belongs to $\mathcal{G}$. Under the directed triangle inequality,*

$$k_{\min}^{\text{WL}} \leq L_{\text{match}} \leq \lceil D_{\text{sep}} \rceil \leq \lceil D_{\text{mix}} \rceil \leq \lceil D_{\text{rob}} \rceil \leq L_{\text{MBP}} \leq k_{\min}^*. \quad (60)$$

Status: rigorously established under the directed triangle inequality.

Proof. The identity function recovers L_{match} . The inequality $D_{\text{sep}} \leq D_{\text{mix}}$ follows because Δ_p contains every unit vector, and $D_{\text{mix}} \leq D_{\text{rob}}$ is the standard max–min inequality. For every fixed matching M , any packing of M into k bins satisfies

$$C_r(M) \leq k \quad r = 1, \dots, p,$$

because each g^r is dual feasible. Thus $\max_r C_r(M) \leq \text{BP}(M)$, and minimizing over M gives $\lceil D_{\text{rob}} \rceil \leq L_{\text{MBP}}$ by (45). Finally, every physical route solution induces a feasible matching and bin assignment by Theorem 7, so $L_{\text{MBP}} \leq k_{\min}^*$. $\square$

The separate bounds require p typewise assignment solves, while D_{mix} is computable by one polynomial-size LP. The robust value D_{rob} is a compact matching MILP; no claim of polynomial-time solvability is made for it. In a fixed- K matching master, all inequalities

$$\sum_{h \in \mathcal{H}} \sum_{(i,j) \in E_h} g^r(s_{ij}) y_{ij} \leq K, \quad r = 1, \dots, p, \quad (61)$$

can be imposed simultaneously before invoking any bin-packing oracle.

Status: a plausible research direction.

The optimized-mixture bound $\lceil D_{\text{mix}} \rceil$ is the most attractive new DFF-based relaxation to test: it is stronger than optimizing the library members independently, remains polynomially computable, and directly strengthens the matching master used to compute L_{MBP} .

Every physical route solution maps to a feasible matching and bin assignment. Conversely, the MBP model omits action sequencing, state transitions, and connector travel, so it is a relaxation rather than an exact formulation. It is NP-hard because it contains bin packing, but it is substantially smaller and more structured than the complete SPDP.

The MBP bound and the duration-IP bound are not generally comparable. Their maximum should be used.

8 Exact matching-based logic Benders algorithm

For a selected pair $p = (i, j)$, interpret p as a fixed token job

$$C_i^p \longrightarrow \beta_i \longrightarrow C_j^d.$$

The matching-bin-packing model is used as the master problem. A route checker determines whether the fixed pairs assigned to each bin can be sequenced on one capacity-two route within duration T .

```

Require: SPDP instance, valid lower bound L, feasible upper bound U
Ensure:  kmin* and routes, or a proof of infeasibility up to U

Build singleton-feasible matching graphs using dij ≤ T
if some type graph has no perfect matching then
    return globally infeasible
Strengthen L using service, assignment, DFF, duration, and MBP bounds
C ← empty set of route-infeasibility cuts

for k = L, L+1, ..., U do
    repeat
        Solve MBP feasibility with at most k bins and cuts C
        if the master is infeasible then
            break
        allFeasible ← true
        for each used bin b, in parallel, do
            Jb ← fixed pickup-delivery pairs assigned to b
            (feasibleb, routeb) ← ExactSingleRouteDP(Jb)
            if feasibleb = false then
                allFeasible ← false
                Cb ← an inclusion-minimal infeasible subset of Jb
                Add  $\sum_{p \in C_b} x_{\{p, b'\}} \leq |C_b| - 1$ 
                    for every bin label b'
        if allFeasible = true then
            return k and the recovered routes
    until the master is infeasible
return no solution with at most U routes

```

8.1 Exact route-checking dynamic program

For r fixed pairs, use identity-expanded states $F(i)$ and $E(i)$ and define

$$\text{dp}[M, v, \sigma] = \text{minimum elapsed duration of a route prefix serving mask } M \text{ and ending at } (v, \sigma). \quad (62)$$

Here M has $2r$ bits, one for every selected pick-up and delivery action. Compatible state-expanded edges extend a label to an unvisited action. A complete mask ending in the empty state is accepted if its duration is at most T .

One duration label per (M, v, σ) is sufficient because all future feasibility depends only on the current state and elapsed duration. The worst-case time and space are

$$O(2^{2r} |\mathcal{E}_r|) = O(4^r \text{poly}(r)). \quad (63)$$

The checker must enforce the exact master pairing $i \mapsto j$. The compact state $E(h)$ erases this identity and is therefore insufficient for fixed-pair infeasible-core cuts. A small identity-expanded checker is needed locally.

8.2 Correctness and complexity

Theorem 8 (Correctness of Algorithm 1). *If every single-route subproblem is solved exactly, Algorithm 1 returns the exact value $k_{\min}^*$ whenever $U \geq k_{\min}^*$.*

Status: rigorously established under the directed triangle inequality.

Proof. Every physical solution with at most k routes induces a feasible master matching and bin assignment. If all route checks pass, their union covers every pick-up and delivery once and is a feasible SPDP solution.

For cut validity, suppose a fixed-pair set C is route-infeasible. If a route contained C together with additional pairs, deleting the additional token lifecycles and shortcutting would produce a feasible route for C , a contradiction. Thus each core cut is valid.

There are finitely many typewise matchings and bin assignments. Each failed assignment produces a cut that removes it, so the algorithm eventually finds a feasible route system or proves the candidate k infeasible. Starting from a valid lower bound, the first successful k is $k_{\min}^*$. $\square$

A crude upper bound on the number of matching/bin assignments for a fixed k is

$$k^n \prod_{h \in \mathcal{H}} |\mathcal{I}_h|!, \quad (64)$$

so exponential worst-case behavior remains. This is unavoidable by Theorem 3.

8.3 Practical variants

Status: a plausible research direction.

The implementation should test $k = L$ first because the benchmark evidence suggests that this candidate is usually feasible. If it fails and $U - L$ is large, one can switch to binary search using the at-most- k decision oracle. Additional improvements include:

- exact two-pair incompatibility cuts generated lazily;
- reuse of Benders cuts across fleet-size candidates;
- warm starts from the matching induced by the duration solution;
- parallel single-route checks;
- minimum-duration edge dominance and prefix-suffix pruning; and
- greedy deletion to extract inclusion-minimal infeasible cores.

A time-limited version is a structured heuristic. If it constructs a feasible solution at a certified lower bound, the solution is exact. No general approximation guarantee should be claimed. In the fixed-matching star subclass, ordinary bin-packing guarantees transfer, but they do not extend automatically to the full SPDP.

9 Polynomial and parameterized special cases

9.1 All physical locations co-located

Proposition 12. *Suppose all pick-up, treatment, and delivery locations coincide, travel times are zero, and $S > 0$. Let $q = \lfloor T/S \rfloor$. If $q \geq 1$, then*

$$k_{\min}^* = \left\lceil \frac{n}{q} \right\rceil. \quad (65)$$

Status: rigorously established from the manuscript’s assumptions.

Every route performs at most q token lifecycles, and any collection of at most q requests can be served sequentially.

9.2 At most one matched token per route

Proposition 13. *If $2S > T$, every route contains at most one matched token. Feasibility is characterized by the perfect matchings in Theorem 5; if the instance is feasible, then $k_{\min}^* = n$.*

Status: rigorously established under the directed triangle inequality.

9.3 At most two fixed jobs per route

Proposition 14. *Suppose the token matching is fixed, every singleton job is feasible, and no route containing three jobs is feasible. Let G be the compatibility graph in which two jobs are*

adjacent if they fit together on one route. Then

$$k_{\min}^* = n - \nu(G), \quad (66)$$

where $\nu(G)$ is the cardinality of a maximum matching.

Status: rigorously established under the directed triangle inequality.

9.4 Exact exponential oracle for small instances

Status: rigorously established from the manuscript's assumptions.

A Held–Karp dynamic program on (action mask, v, σ) can enumerate feasible route masks in $O(4^n \text{poly}(n))$ time. A subsequent exact-cover dynamic program requires $O(9^n)$ time and $O(4^n \text{poly}(n))$ space. This is fixed-parameter tractable in n and is useful as an independent small-instance oracle.

The problem is not fixed-parameter tractable in k alone unless $P = NP$. Nor is it fixed-parameter tractable in maximum route cardinality alone, because the 3-PARTITION construction remains hard when each route contains at most three requests.

10 A rigorous reformulation of Section 6

Define

$$\phi_{\text{IP}}(k) = \min\{C(y) : (y, B) \in \mathcal{F}^{\text{IP}}, K(y) = k\}, \quad (67)$$

using $+\infty$ for infeasible fleet sizes. Then

$$z_{\text{IP}}(f) = \min_{k \geq k_{\min}^*} \{fk + \phi_{\text{IP}}(k)\}. \quad (68)$$

10.1 Exact fixed-charge threshold

Theorem 9 (Fleet-size threshold). *Let $k_0 = k_{\min}^*$. A minimum-fleet solution is optimal for the original objective if and only if*

$$f \geq f_{\text{crit}} := \max_{\substack{k > k_0 \\ \phi_{\text{IP}}(k) < +\infty}} \frac{\phi_{\text{IP}}(k_0) - \phi_{\text{IP}}(k)}{k - k_0}, \quad (69)$$

where the maximum is truncated below at zero. Strict inequalities give uniqueness of the optimal fleet size.

Status: rigorously established from the manuscript's assumptions.

A simple sufficient condition is

$$f > \Delta_{\text{IP}}(k_0), \quad (70)$$

because every solution with more vehicles has variable cost at least $c_{\min}$. A computable sufficient certificate is

$$f > \bar{C}_{k_0} - \underline{c}, \quad (71)$$

where $\bar{C}_{k_0}$ is the variable cost of any feasible k_0 -route solution and $\underline{c}$ is a valid global lower bound on travel cost.

10.2 Post-fleet-cut integrality gap

Let

$$z_{\text{LP}}^+(f) = \min\{C(y) + fK(y) : (y, B) \in \mathcal{F}^{\text{LP}}, K(y) \geq k_0\}. \quad (72)$$

Theorem 10 (Conditional post-fleet-cut gap). *Suppose that an original-cost integer optimum uses $k_0 = k_{\min}^*$ and that*

$$\phi_{\text{LP}}(k_0) \leq c_{\min}. \quad (73)$$

Then

$$z_{\text{IP}}(f) - z_{\text{LP}}^+(f) \geq \Delta_{\text{IP}}(k_0). \quad (74)$$

Status: rigorously established from the manuscript's assumptions.

Indeed,

$$z_{\text{IP}}(f) = fk_0 + c_{\min} + \Delta_{\text{IP}}(k_0),$$

whereas the assumed fixed- k_0 LP solution gives

$$z_{\text{LP}}^+(f) \leq fk_0 + \phi_{\text{LP}}(k_0) \leq fk_0 + c_{\min}.$$

Condition (73) must be verified computationally; it is not automatic.

10.3 Conditional fleet-profile inequality

Let $\underline{\phi}_0$ be any valid lower bound on $\phi_{\text{IP}}(k_0)$ and let $\underline{c} \leq c_{\min}$. Define

$$\hat{\Delta} = \max\{0, \underline{\phi}_0 - \underline{c}\}. \quad (75)$$

Proposition 15 (Fleet-profile cut). *The inequality*

$$C(y) + \hat{\Delta}(K(y) - k_0) \geq \underline{\phi}_0 \quad (76)$$

is valid for every integer solution satisfying $K(y) \geq k_0$.

Status: rigorously established from the manuscript's assumptions.

At $K = k_0$, the inequality imposes $C(y) \geq \underline{\phi}_0$. At $K \geq k_0 + 1$, its right-hand-side requirement on $C(y)$ is no larger than $\underline{c}$, which is globally valid.

For the exact A8 and A9 values reported in the manuscript, the corresponding inequalities are

$$C(y) + 13(K(y) - 2) \geq 314, \quad (77)$$

$$C(y) + (K(y) - 2) \geq 244. \quad (78)$$

These cuts directly target the residual conditional gaps identified in Section 6. More generally, the lower convex envelope of certified fleet-cost profile points produces a family of aggregate

(K, C) epigraph inequalities.

11 Recommended contribution hierarchy

Table 1: Preliminary assessment of proposed contributions.

Contribution	Novelty	Feasibility	Likely impact
Matching-workload theorem, assignment/DFP hierarchy, and MBP bound	High	Very high	High
Matching-based logic Benders certification	High	Medium	Very high if effective
Conditional fleet-profile inequalities	Medium-high	High	High
Fleet-refinement and matching-feasibility theorems	Medium	Very high	Medium-high
Hardness and duration-bound counterexamples	Moderate	Very high	Foundational
Busy-block recombination heuristic	Medium	High	Empirical

Status: a plausible research direction.

The most promising central contribution is a *matching-aware lower-bound and exact certification framework for the minimum fleet size*. At minimum, the paper should establish:

- (i) the routewise matching-workload theorem;
- (ii) the hierarchy $H^{\text{WL}} \leq H^{\text{match}} \leq H^*$;
- (iii) polynomial assignment and dual-feasible-function fleet bounds;
- (iv) fleet-specific graph reductions; and
- (v) the sandwich certification theorem with end-to-end computational evidence.

The logic-based Benders algorithm is the most ambitious extension. The conditional fleet-profile inequality is the strongest way to retain and strengthen the current Section 6 narrative.

12 Computational experiments

Status: a plausible research direction.

12.1 Baselines

The experiments should compare:

- (i) the Wøhlk–Laporte workload bound;
- (ii) the service-cardinality bound (23);
- (iii) the assignment bound L_{match} ;

- (iv) the maximum over several dual-feasible-function bounds;
- (v) the exact MBP lower bound;
- (vi) duration-LP and duration-IP bounds;
- (vii) the fragment-LP fleet bound;
- (viii) direct minimization of K in the 2-index model;
- (ix) sequential fixed- k feasibility;
- (x) at-most- k binary search;
- (xi) the proposed logic-based Benders method; and
- (xii) the subset-DP oracle on small instances.

12.2 Instance classes

In addition to the A–D benchmark instances, the test set should include:

- star/bin-packing instances with token workloads near $T/2$ and $T/3$;
- Hamiltonian-path-derived sequencing instances;
- geographically separated instances on which $\lceil H^*/T \rceil$ is weak;
- matching-collision instances in which many rows share the same cheapest delivery candidate;
- sequencing-adversarial instances for which MBP is feasible but the corresponding bins are not route-realizable;
- several skip-type multiplicity patterns;
- several duration limits near fleet-size transition thresholds; and
- symmetric and asymmetric metric instances with varying service-to-travel ratios.

12.3 Metrics and robustness

Report the following:

- every lower bound and its gap to the certified $k_{\min}^*$;
- the frequency with which each bound equals $k_{\min}^*$;
- bound-computation time and time to the first feasible L -route solution;
- total end-to-end certification time, including auxiliary solves;
- branch-and-bound nodes, memory, and final gaps;
- Benders iterations, route checks, and infeasible-core sizes;

- the marginal effect of every graph reduction and cut family; and
- robustness over documented hardware, fixed thread counts, and multiple solver seeds.

For Section 6, compute $\phi_{\text{LP}}(k_{\min}^*)$ directly and report

$$\phi_{\text{LP}}(k_{\min}^*), \quad \phi_{\text{IP}}(k_{\min}^*), \quad c_{\min}, \quad \Delta_{\text{IP}}(k_{\min}^*).$$

The fleet-profile inequalities should be tested both at the root and over the complete branch-and-bound tree. Any relationship between Δ_{IP} and node counts should be reported as empirical correlation rather than causation.

13 Recommended reorganization of the manuscript

1. **Section 5.1: Minimum-fleet problem and route exactness.** Introduce the physical and set-partitioning definitions, prove route decomposition, and state the complexity theorem.
2. **Section 5.2: Structural fleet bounds.** Present the service-cardinality bound, matching-feasibility theorem, matching-workload theorem, assignment/DFF/MBP hierarchy, and fleet-specific graph reductions.
3. **Section 5.3: Exact certification and incumbent construction.** State the sandwich theorem, compare sequential and binary search, present the logic-based Benders algorithm, and explain fixed-fleet incumbent recovery.
4. **Section 6: Original-objective trade-off.** Retain the busy-block splitting argument, introduce the fleet-cost profile and fixed-charge threshold, prove the conditional-gap theorem, and derive fleet-profile inequalities.
5. **Computational section.** Separate minimum-fleet certification experiments from original-objective experiments, include all auxiliary computation in end-to-end timing, and directly evaluate the equality-constrained LP quantities used in Section 6.

The revised narrative should be that minimum-fleet certification is worst-case hard, but the flexible SPDP contains a latent matching-and-packing structure. Exploiting this structure produces strong lower bounds and short exact certificates on the benchmark instances. The original fixed-charge objective can remain difficult because its LP relaxation does not represent the conditional travel-cost profile at the minimum feasible fleet size.